

PROJECT 1

Group Project Task List

Student Name	Student Id	Role
Tessalonika Angeline Purba	001202400210	Team Leader
Qwyn Celine Djimondo	001202400205	Member 1
Tentin Nofa Tuzika	001202400134	Member 2

Chapter 1. Introduction

As a group, identify potential problems in your daily life that can be solved using programming.

“Menstrual Cycle Tracker and Reminder”

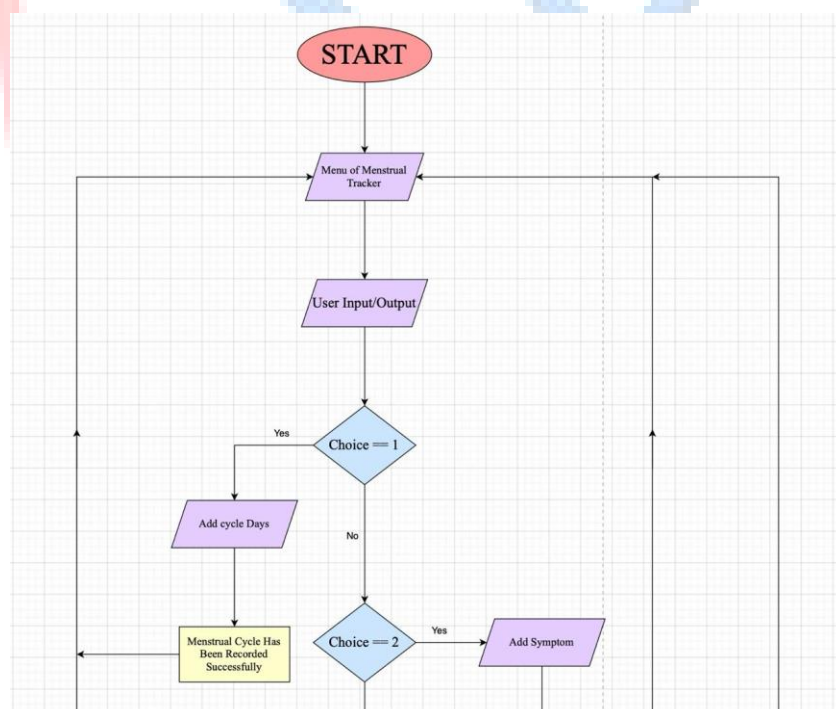
Many women find it difficult to track their menstrual cycles, especially irregular ones. This Menstrual Cycle Tracker and Reminder program aims to help users record how many days their period lasts, predict the next cycle, send reminders before the cycle starts, record symptoms or moods, and read health information related to menstruation.

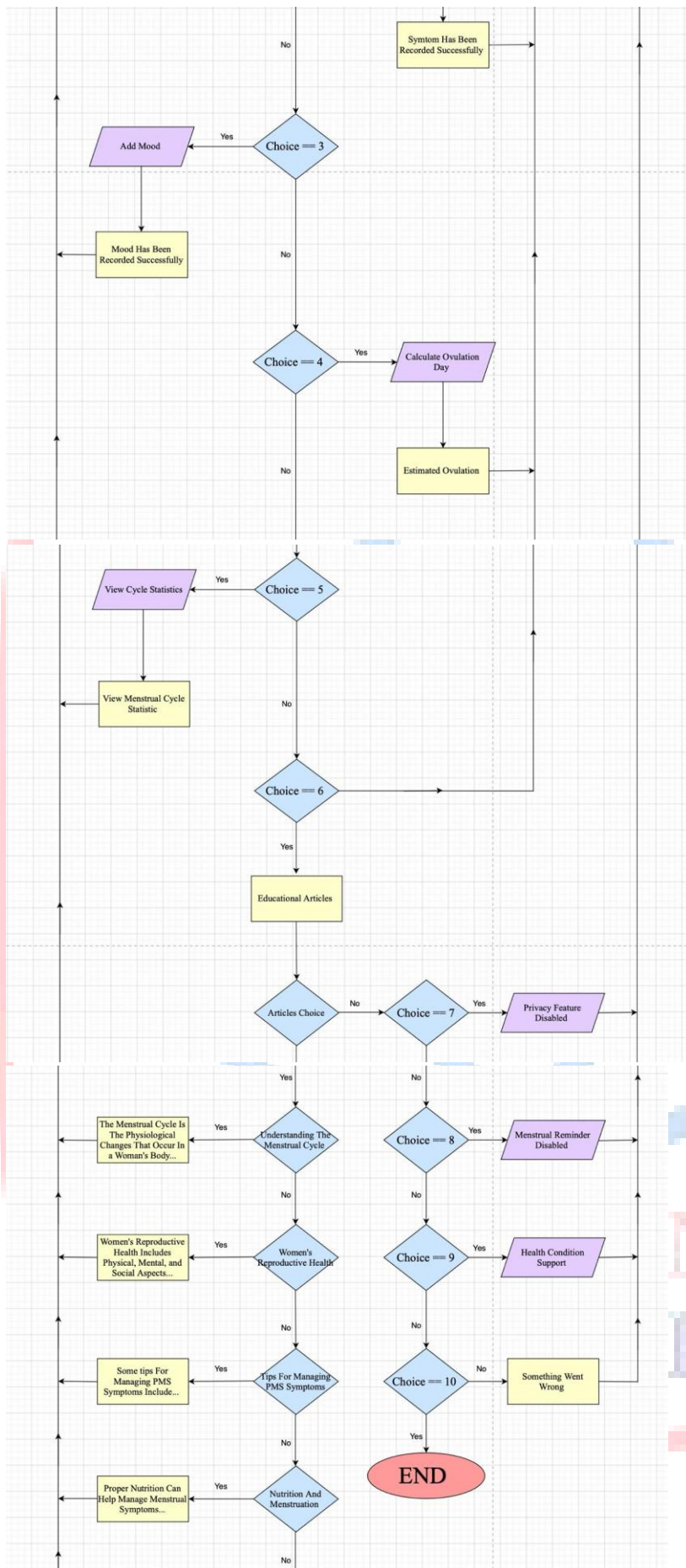
Features:

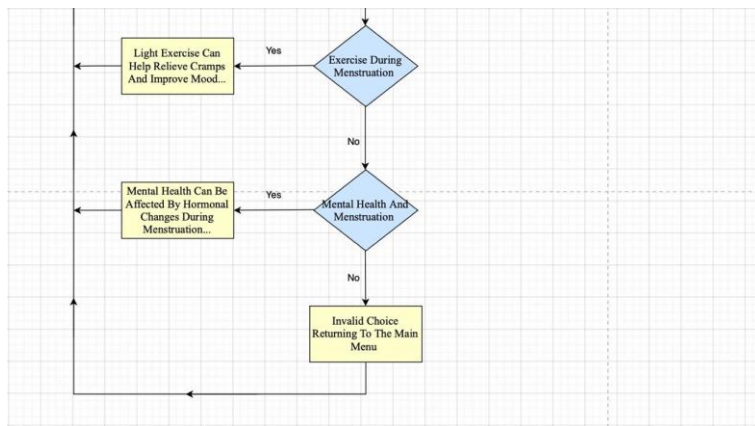
1. Enable users to record menstrual cycles
2. Enable users to add symptoms
3. Enable users to add moods
4. Enable users to calculate ovulation days
5. Enable users to view cycle statistics
6. Enable users to read health education articles
7. Enable users to enable/disable data privacy
8. Enable users to enable/disable menstrual reminders
9. Enable users to read support for health conditions

Chapter 2. Flowchart

on the description you provided in Chapter 1, provide the end-to-end flowchart of your program







Chapter 3: The Program

Provide the program in this section (copy and paste your program here)

Line	Code: “Menstrual Cycle Tracker and Reminder”
1	#include <iostream>
2	#include <string>
3	#include <numeric> // For average calculation
4	#include <iomanip> // For decimal formatting
5	#include <ctime> // For setting reminder dates
6	
7	using namespace std;
8	
9	class MenstrualTracker {
10	private:
11	static const int MAX_RECORDS = 100; // Maximum number of records
12	
13	int cycleDays[MAX_RECORDS]; // Stores the duration of menstrual cycles
14	string symptoms[MAX_RECORDS]; // Stores symptoms experienced
15	string moods[MAX_RECORDS]; // Stores mood states
16	
17	int cycleCount; // Tracks the number of recorded cycles
18	int symptomCount; // Tracks the number of recorded symptoms
19	int moodCount; // Tracks the number of recorded moods
20	
21	int lastCycleDuration; // Duration of the last cycle
22	bool dataPrivacyEnabled; // Simulates privacy feature
23	bool reminderEnabled; // Simulates menstrual reminders
24	
25	public:
26	MenstrualTracker()
27	: cycleCount(0), symptomCount(0), moodCount(0),
28	lastCycleDuration(28), dataPrivacyEnabled(true), reminderEnabled(true) {}
29	
30	// Function to record menstrual cycle
31	void trackCycle(int days) {

```

32     if (cycleCount < MAX_RECORDS) {
33         cycleDays[cycleCount++] = days;
34         lastCycleDuration = days;
35         cout << "Menstrual cycle of " << days << " days has been recorded
36 successfully.\n";
37     } else {
38         cout << "Cycle records are full. Unable to record more.\n";
39     }
40 }
41
42 // Function to add symptoms experienced by the user
43 void addSymptom(const string& symptom) {
44     if (symptomCount < MAX_RECORDS) {
45         symptoms[symptomCount++] = symptom;
46         cout << "Symptom \"\" << symptom << "\" has been recorded
47 successfully.\n";
48     } else {
49         cout << "Symptom records are full. Unable to record more.\n";
50     }
51 }
52
53 // Function to add the user's mood
54 void addMood(const string& mood) {
55     if (moodCount < MAX_RECORDS) {
56         moods[moodCount++] = mood;
57         cout << "Mood \"\" << mood << "\" has been recorded successfully.\n";
58     } else {
59         cout << "Mood records are full. Unable to record more.\n";
60     }
61 }
62
63 // Function to calculate ovulation day
64 void calculateOvulation() {
65     if (cycleCount == 0) {
66         cout << "No cycle data recorded yet.\n";
67         return;
68     }
69
70     int totalCycleDays = 0;
71     for (int i = 0; i < cycleCount; i++) {
72         totalCycleDays += cycleDays[i];
73     }
74
75     int averageCycle = totalCycleDays / cycleCount;
76     int ovulationDay = averageCycle - 14;
77     cout << "Estimated ovulation day: Day " << ovulationDay << " of the cycle.\n";
78 }
79
80 // Function to display basic menstrual cycle statistics
81 void displayStatistics() {
82     if (cycleCount == 0) {
83         cout << "No statistics data to display yet.\n";
84         return;
85     }
86
87     int totalDays = 0;

```

```

88     int maxCycle = cycleDays[0];
89     int minCycle = cycleDays[0];
90
91     for (int i = 0; i < cycleCount; i++) {
92         totalDays += cycleDays[i];
93         if (cycleDays[i] > maxCycle) maxCycle = cycleDays[i];
94         if (cycleDays[i] < minCycle) minCycle = cycleDays[i];
95     }
96
97     double averageCycle = static_cast<double>(totalDays) / cycleCount;
98
99     cout << "\n--- Menstrual Cycle Statistics ---\n";
100    cout << "Number of recorded cycles: " << cycleCount << "\n";
101    cout << "Average cycle duration: " << fixed << setprecision(2) << averageCycle
102    << " days.\n";
103    cout << "Longest cycle duration: " << maxCycle << " days.\n";
104    cout << "Shortest cycle duration: " << minCycle << " days.\n";
105    }
106
107    // Function to display educational health articles (simulation)
108    void displayEducationalArticles() {
109        cout << "\n--- Educational Health Articles ---\n";
110        cout << "1. Understanding the Menstrual Cycle\n";
111        cout << "2. Women's Reproductive Health\n";
112        cout << "3. Tips for Managing PMS Symptoms\n";
113        cout << "4. Nutrition and Menstruation\n";
114        cout << "5. Exercise During Menstruation\n";
115        cout << "6. Mental Health and Menstruation\n";
116        cout << "Enter article number to read more (0 to go back): ";
117
118        int articleChoice;
119        cin >> articleChoice;
120
121        switch (articleChoice) {
122            case 1:
123                cout << "\n--- Understanding the Menstrual Cycle ---\n";
124                cout << "The menstrual cycle is the physiological changes that occur in a
125                woman's body...\n";
126                break;
127            case 2:
128                cout << "\n--- Women's Reproductive Health ---\n";
129                cout << "Women's reproductive health includes physical, mental, and
130                social aspects...\n";
131                break;
132            case 3:
133                cout << "\n--- Tips for Managing PMS Symptoms ---\n";
134                cout << "Some tips for managing PMS symptoms include...\n";
135                break;
136            case 4:
137                cout << "\n--- Nutrition and Menstruation ---\n";
138                cout << "Proper nutrition can help manage menstrual symptoms...\n";
139                break;
140            case 5:
141                cout << "\n--- Exercise During Menstruation ---\n";
142                cout << "Light exercise can help relieve cramps and improve mood...\n";
143                break;
144

```

```

145         case 6:
146             cout << "\n--- Mental Health and Menstruation ---\n";
147             cout << "Mental health can be affected by hormonal changes during
148 menstruation...\n";
149             break;
150         case 0:
151             cout << "Returning to the main menu...\n";
152             break;
153         default:
154             cout << "Invalid choice. Returning to the main menu.\n";
155             break;
156     }
157 }
158
159 // Function to enable or disable the privacy feature
160 void toggleDataPrivacy() {
161     dataPrivacyEnabled = !dataPrivacyEnabled;
162     cout << "Privacy feature " << (dataPrivacyEnabled ? "enabled.\n" :
163 "disabled.\n");
164 }
165
166 // Function to enable or disable menstrual reminders
167 void toggleReminder() {
168     reminderEnabled = !reminderEnabled;
169     cout << "Menstrual reminder " << (reminderEnabled ? "enabled.\n" :
170 "disabled.\n");
171 }
172
173 // Function to display health condition support (simulation)
174 void supportHealthConditions() {
175     cout << "\n--- Health Condition Support ---\n";
176     cout << "1. PCOS (Polycystic Ovary Syndrome)\n";
177     cout << "2. Endometriosis\n";
178     cout << "3. Hormonal Imbalance\n";
179 }
180
181 // Simple user interface for selecting features
182 void userFriendlyInterface() {
183     cout << "\n--- Menstrual Tracker ---\n";
184     cout << "1. Record menstrual cycle\n";
185     cout << "2. Add symptom\n";
186     cout << "3. Add mood\n";
187     cout << "4. Calculate ovulation day\n";
188     cout << "5. View cycle statistics\n";
189     cout << "6. Read educational health articles\n";
190     cout << "7. Enable/disable data privacy\n";
191     cout << "8. Enable/disable menstrual reminders\n";
192     cout << "9. Support for health conditions\n";
193     cout << "10. Exit\n";
194 }
195
196 // Function to send menstrual reminders if enabled
197 void sendReminder() {
198     if (reminderEnabled) {
199         cout << "\n[Reminder] Your menstrual cycle may start in a few days. Please
200 be prepared.\n";

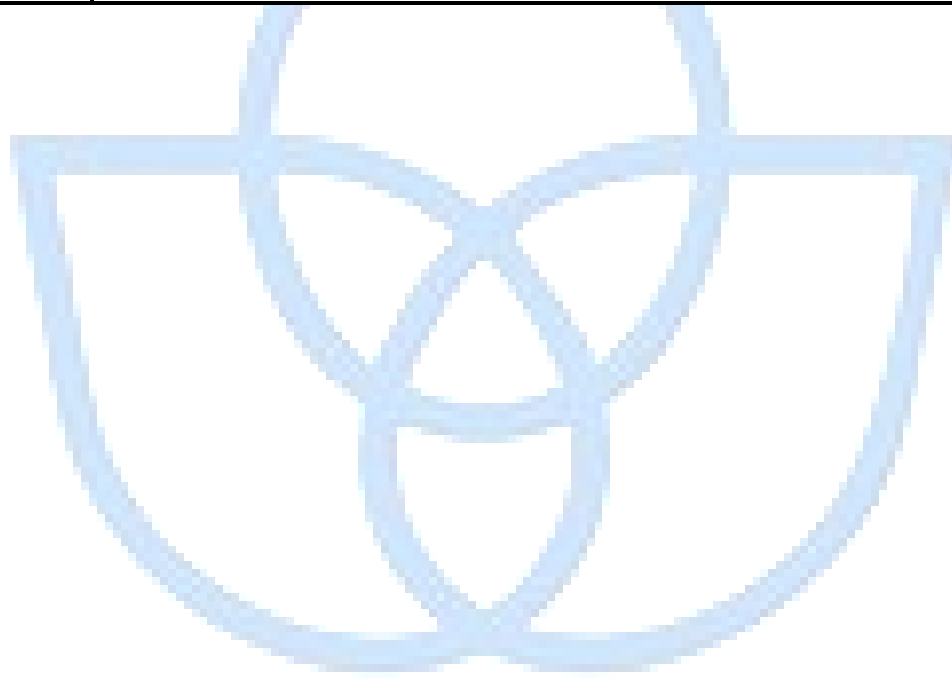
```

```

201     }
202     }
203 };
204
205 int main() {
206     MenstrualTracker tracker;
207     int choice;
208     int cycleDays;
209     string symptom, mood;
210
211     do {
212         tracker.userFriendlyInterface();
213         cout << "Choose an option: ";
214         cin >> choice;
215
216         switch (choice) {
217             case 1: // Record menstrual cycle
218                 cout << "Enter the duration of the menstrual cycle (days): ";
219                 cin >> cycleDays;
220                 tracker.trackCycle(cycleDays);
221                 tracker.sendReminder(); // Send reminder if enabled
222                 break;
223             case 2: // Add symptom
224                 cout << "Enter symptom: ";
225                 cin.ignore(); // Clear buffer
226                 getline(cin, symptom);
227                 tracker.addSymptom(symptom);
228                 break;
229             case 3: // Add mood
230                 cout << "Enter mood: ";
231                 cin.ignore(); // Clear buffer
232                 getline(cin, mood);
233                 tracker.addMood(mood);
234                 break;
235             case 4: // Calculate ovulation day
236                 tracker.calculateOvulation();
237                 break;
238             case 5: // View cycle statistics
239                 tracker.displayStatistics();
240                 break;
241             case 6: // Read educational health articles
242                 tracker.displayEducationalArticles();
243                 break;
244             case 7: // Enable/disable data privacy
245                 tracker.toggleDataPrivacy();
246                 break;
247             case 8: // Enable/disable menstrual reminders
248                 tracker.toggleReminder();
249                 break;
250             case 9: // Support for health conditions
251                 tracker.supportHealthConditions();
252                 break;
253             case 10: // Exit
254                 cout << "Exiting the program.\n";
255                 break;
256             default:

```

258	cout << "Invalid choice. Please try again.\n";
259	}
260	
261	} while (choice != 10);
262	
263	return 0;
264	}
265	
266	
267	
268	
269	
270	
271	
272	
273	
274	
275	
276	



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Chapter 4: The Screenshot

Give a screenshot of your feature here. Each screenshot should represent 1 feature, as listed in Chapter 1.

1st Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 1
Enter the duration of the menstrual cycle (days): 7
Menstrual cycle of 7 days has been recorded successfully.

[Reminder] Your menstrual cycle may start in a few days. Please be prepared.
```

2nd Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 2
Enter symptom: stomach ache and back pain
Symptom "stomach ache and back pain" has been recorded successfully.
```

3rd Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 3
Enter mood: hungry and sick
Mood "hungry and sick" has been recorded successfully.
```

4th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 4
Estimated ovulation day: Day -7 of the cycle.
```

5th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 5
|
--- Menstrual Cycle Statistics ---
Number of recorded cycles: 1
Average cycle duration: 7.00 days.
Longest cycle duration: 7 days.
Shortest cycle duration: 7 days.
```

6th Feature Code

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit

Choose an option: 6

--- Educational Health Articles ---

1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

Enter article number to read more (0 to go back): 1

--- Understanding the Menstrual Cycle ---

The menstrual cycle is the physiological changes that occur in a woman's body...

7th Feature Code

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit

Choose an option: 6

--- Educational Health Articles ---

1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

Enter article number to read more (0 to go back): 2

--- Women's Reproductive Health ---

Women's reproductive health includes physical, mental, and social aspects...

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit

Choose an option: 6

--- Educational Health Articles ---

1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

Enter article number to read more (0 to go back): 3

--- Tips for Managing PMS Symptoms ---

Some tips for managing PMS symptoms include...

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit

Choose an option: 6

--- Educational Health Articles ---

1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

Enter article number to read more (0 to go back): 4

--- Nutrition and Menstruation ---

Proper nutrition can help manage menstrual symptoms...

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit

Choose an option: 6

--- Educational Health Articles ---

1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

Enter article number to read more (0 to go back): 5

--- Exercise During Menstruation ---

Light exercise can help relieve cramps and improve mood...

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit

Choose an option: 6

--- Educational Health Articles ---

1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

Enter article number to read more (0 to go back): 6

--- Mental Health and Menstruation ---

Mental health can be affected by hormonal changes during menstruation...

12th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 7
Privacy feature disabled.
```

13th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 8
Menstrual reminder disabled.
```

14th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 9

--- Health Condition Support ---
1. PCOS (Polycystic Ovary Syndrome)
2. Endometriosis
3. Hormonal Imbalance
```

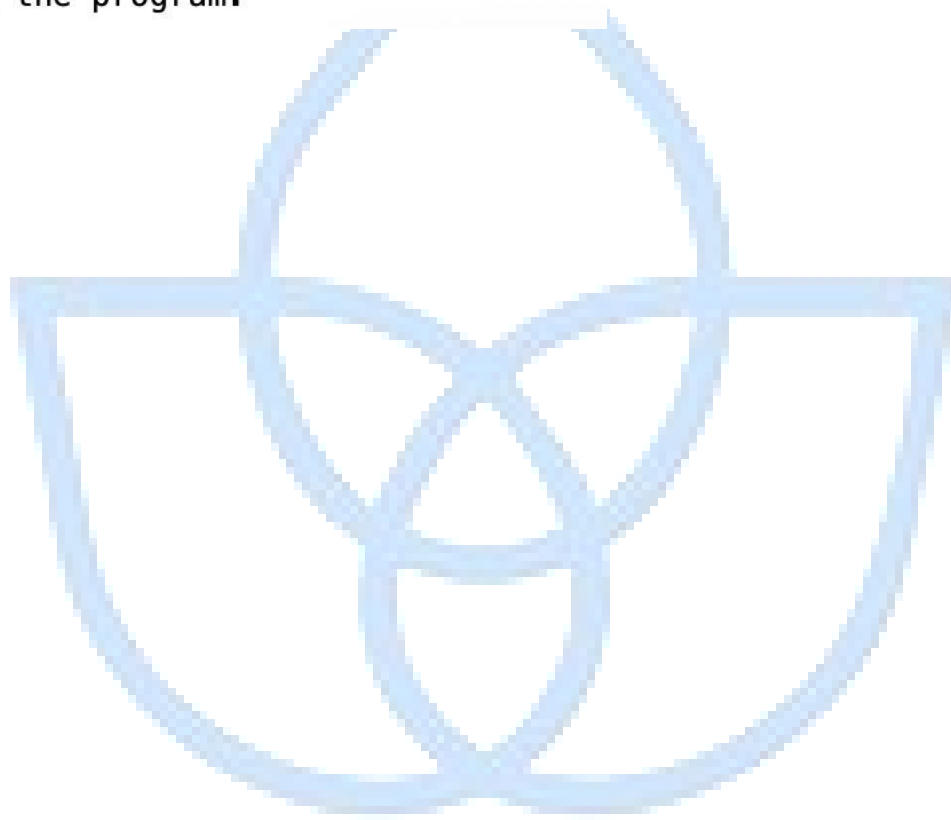
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--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit

Choose an option: 10

Exiting the program.



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Chapter 5: Requirement checks

No	Requirements	Yes/No, Line No X - Y
1	Have at least one selection.	Yes, If-else Line 32-38 Switch Line 118-149, 206-248
2	Have at least one loop.	Yes, Loop Line 201-250
3	Have at least one 1 Dimensional Array.	Yes, 1 Dimensional Array Line 13-15
4	Have at least one mathematical operator calculation.	Yes, 1 Mathematical Operation Calculation Line 68-76, 85-95
5	Have at least one Boolean expression.	Yes, Boolean Expression Line 22, 23, 153-156, 159-162

Chapter 6: Log Hours

Student Name	Activity	Log Hours
Tessalonika Angeline Purba	- Discussion/Planning - Report/Documentation - Flowchart - Code first and second feature (1-Dimensional Array statement, Switch statement, User input, Mathematical operator calculation, Print output)	- 1 Hour - 2 Days - 20 minutes - 1 hour
Qwyn Celine Djimondo	- Discussion/Planning - Report/Documentation - Flowchart - Code first and second feature (1-Dimensional Array statement, Switch statement, User input, Mathematical operator calculation, Print output)	- 1 Hour - 10 minutes - 2 Days - 30 minutes
Tentin Nofa Tuzika	- Discussion/Planning - Report/Documentation - Flowchart - Code first and second feature (1-Dimensional Array statement, Switch statement, User input, Mathematical operator calculation, Print output)	- 1 Hour - 10 minutes - 1 Days - 45 minutes